

CareerAutomated Built an autonomous 10-Agent AI platform that finds jobs, researches companies, tailors LaTeX resumes, and submits applications end-to-end with zero human intervention. Implemented component-based architecture separating discovery, intelligence, routing, and execution. Designed human-in-the-loop review gate for safety before final submission. Developed waterfall UI normalization strategy (Exact -> Rules -> Fuzzy) to handle modern custom ATS widgets like React-Select. Built embedding-free RAG using BM25 for zero-cost, high-speed knowledge retrieval. Implemented strict prompt grounding rules and low-confidence gating to prevent LLM hallucinations on business metrics.

AI Data Analyst Agent Designed a 5-agent LangGraph system where non-technical users type a question in plain English and get a data analysis result in under 60 seconds — no SQL, no engineer required. Implemented LangGraph for stateful agent workflow (not vanilla LangChain) — explicit state management per agent. Query Planner does intent classification BEFORE generation — eliminates downstream type errors. Used n8n as orchestration layer separate from agent logic — non-technical operators can modify routing without redeployment. Implemented MCP to expose tools (files, DBs, APIs) as standardised context — extensible by design. Designed Error Fixer agent: classify exception → regenerate targeted fix → retry 3x before user escalation (ReAct-style). Used Docker sandbox to isolate LLM-generated code from host.

Backend Systems Deployed FastAPI + Docker + AWS EC2 for scalable backend infrastructure. Implemented SQL for analytical queries (funnel, cohort, window functions). Used Python for data and backend development.

OrangeLabs Internship Contributed to CCTV-based automated attendance (computer vision pipeline → ERP sync). Developed AI-generated personalised lecture pipeline (teacher voice/likeness cloning for student-facing content). Implemented n8n ERP workflow automation (weaker signal, de-emphasised).